Gantry Motor and Driver Recommendations

COMP 490/L | Research summary for team review | November 5, 2026

Goal: Four open-loop motors for X/Y movement (two per axis) on a small belt-driven gantry. We prefer independent motor channels, good documentation, reasonable cost, and precision over speed. The 10 lb prototype payload is a goal, not a verified motor-sizing specification.

Ranked options

Rank	Motor + driver	4-motor cost*	Main reason / concern
1	17HS15-1504S1 + 2 ST X-NUCLEO-IHM02A1 boards	$68.26-$68.56	Best value. Each ST board drives 2 motors independently. Check board stacking and pin use.
2	17HS19-2004S1 + 2 ST X-NUCLEO-IHM02A1 boards	$70.22-$70.52	Stronger NEMA 17 motor. Slightly larger/heavier; still needs feedback sensors.
3	17HS15-1504S1 + TMCM-3216 + 1 extra driver	$276+	True 3-axis controller/driver plus fourth channel. Encoder inputs, but expensive and harder to integrate.

*Estimated motors + driver boards only, based on manufacturer listings checked Nov 5. Excludes Nucleo board, power supply, encoders, switches, connectors, belts, shipping and tax. Three-motor cost: about $59.13 for option 1, $60.60 for option 2, and $253.94 for option 3 (without the fourth driver).

My recommendation

Start by testing option 1: four StepperOnline 17HS15-1504S1 motors with two ST X-NUCLEO-IHM02A1 boards. The motors are NEMA 17, 45 N·cm holding torque and 1.5 A/phase. Each ST expansion board contains two independent L6470 stepper-driver channels and uses SPI for control. This meets the preference for at least two motors per physical driver board.

If the final motion calculations show that 45 N·cm is insufficient, option 2 uses the higher-torque 17HS19-2004S1 (59 N·cm, 2 A/phase). Option 3 only makes sense if having three independent motors on one board is worth the higher cost.

Quick technical notes

- Option 1 motor: 42 x 42 x 39 mm, 1.8°/step, 45 N·cm holding torque, 1.5 A/phase, 2.3 Ω, 4.4 mH. Listed motor price: $9.13 each.
- Option 2 motor: 42 x 42 x 48 mm, 1.8°/step, 59 N·cm holding torque, 2.0 A/phase, 1.6 Ω, 3.0 mH. Listed motor price: $9.62 each.
- ST X-NUCLEO-IHM02A1: two independent bipolar stepper channels (L6470), 8-45 V motor supply, up to 3 A RMS/channel, 3.3 V logic option, SPI control. Listed price: about $15.87-$16.02 per board.
- Analog Devices TMCM-3216: three independent motor channels, 10-30 V input, 1.5 A RMS/channel, encoder inputs, RS-485/CAN/USB. Listed at approximately $226.55 before adding a fourth driver channel.

Position feedback and safety

Open-loop step commands do not prove that the gantry moved the expected distance. The STM32 can compare commanded position with a motor encoder or carriage-position sensor, report a mismatch, and stop/re-home before attempting correction. A motor encoder can detect missed motor steps, but may not detect belt slip; carriage feedback is better for that. Two motors on one axis also need synchronization and a plan to avoid racking.

For Sprint 2, one motor and one channel of the dual-axis ST board could be used for a small, unloaded homing/movement test. A real emergency stop must independently disable hazardous motor motion; a software stop alone is not sufficient.

Before purchasing

Confirm moving mass, belt pitch, pulley diameter, required acceleration/speed, actual torque at operating speed, supply sizing, sensor choice, and which STM32 board is being used. Verify the chosen Nucleo board can work with two stacked ST driver boards without pin or software conflicts. The 10 lb payload target alone cannot establish motor capacity.

Product pages / documentation

StepperOnline 17HS15-1504S1 (datasheet and torque curve on page)
StepperOnline 17HS19-2004S1 (datasheet and torque curve on page)
ST X-NUCLEO-IHM02A1 technical information
ST X-NUCLEO-IHM02A1 store / price
ST X-CUBE-SPN2 software
Analog Devices TMCM-3216 technical information
Analog Devices TMCM-3216 datasheet